

On the properties of arithmetic crosscorrelation for sequences with coprime periods

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Abstract

Arithmetic correlation is a critical performance measure for pseudorandom sequences generated by feedback with carry shift registers (FCSR), extending classical correlation by accounting for carry propagation. Chen et al. proved that for binary sequences with coprime periods, the arithmetic crosscorrelation is constant, and established bounds for Legendre sequences and m -sequences. In this paper, we further investigate the arithmetic crosscorrelation of sequences with coprime periods. We derive upper bounds for sequences constructed from the Legendre symbol, which generalize classical Legendre sequences, and for sequences generated by trace functions. In addition, we show that the constant property of arithmetic crosscorrelation extends to non-binary sequences with coprime periods.

Keywords: Pseudorandom sequence, Arithmetic crosscorrelations, Legendre symbol, Trace sequences

1 Introduction

Pseudorandom sequences are generated by shift registers and deterministic algorithms, and play a vital role in fields such as communications, radar, and stream ciphers [1, 2]. In the 1990s, Klapper and Goresky introduced the feedback with carry shift register

(FCSR) sequence, which can be viewed as an arithmetic analog of the linear feedback shift register (LFSR) sequence [3].

One important measure for FCSR sequences is arithmetic correlation. The concept of arithmetic autocorrelation was introduced by Mandelbaum [4] in his research on arithmetic codes, and was later extended to arithmetic correlation by Goresky and Klapper in their study of FCSR sequences [2, 5].

For two binary sequences $\mathcal{S} = (s_i)_{i=0}^{N-1}$ and $\mathcal{T} = (t_i)_{i=0}^{N-1}$ with a common minimal period N , where $s_i, t_i \in \{0, 1\}$, their arithmetic crosscorrelation can be defined as follows, according to [2, Proposition 13.3.8] and [6]:

Write

$$S(2) = \sum_{i=0}^{N-1} s_i 2^i, \quad T^{(\tau)}(2) = \sum_{i=0}^{N-1} t_{i+\tau} 2^i.$$

The integer $|S(2) - T^{(\tau)}(2)| \geq 0$ can be expressed in 2-adic expansion $|S(2) - T^{(\tau)}(2)| = \sum_{i=0}^{N-1} w_i 2^i$ ($w_i \in \{0, 1\}$). The arithmetic crosscorrelation of $\mathcal{S}$ and $\mathcal{T}$ at a shift τ is defined as

$$\mathcal{C}_{\mathcal{S}, \mathcal{T}}^A(\tau) = \begin{cases} N_0 - N_1, & \text{if } S(2) \geq T^{(\tau)}(2), \\ N_1 - N_0, & \text{if } S(2) < T^{(\tau)}(2), \end{cases} \quad (1)$$

where N_j is the number of $i = 0, 1, \dots, N-1$ with $w_i = j$ for each $j \in \{0, 1\}$.

Recently, Hofer, Merai, and Winterhof [7] established a relationship between arithmetic autocorrelation and correlation measure of order k , showing that binary sequences with sufficiently small higher order correlation measures cannot exhibit large arithmetic autocorrelation, and applies this result to several families of sequences, including Legendre sequences defined by polynomials and trace function sequences over finite fields. For sequences with coprime periods, Chen et al. [8] proved that their arithmetic crosscorrelation is constant and provided an upper bound for the arithmetic crosscorrelation between Legendre sequences and m -sequences with such periods. Jing et al. [9] later improved the results for the arithmetic crosscorrelation of m -sequences with coprime periods, obtaining the exact value. Related work by Yan et al. [10] established an upper bound estimate for the arithmetic crosscorrelation of Fermat quotient related sequences with coprime periods.

Inspired by the above studies, in Sections 2 and 3 of this paper, we investigate the arithmetic crosscorrelation of sequences constructed from the Legendre symbol and the trace function with coprime periods, respectively, and obtain upper bounds for them. Here, sequences constructed from the Legendre symbol can be regarded as generalizations of Legendre sequences. In Section 4, we prove that the arithmetic crosscorrelation of non-binary sequences with coprime periods is also a constant.

In this paper, we say that $h(n) = O(l(n))$ (also written as $h(n) \ll l(n)$) if there exists an absolute constant $C > 0$ such that $|h(n)| \leq Cl(n)$ for all n . For sequence $\mathcal{S} = (s_i)_{i \geq 0}$ with period N , define $s_{-i} = s_{N-i}$. And $\log$ denotes the natural logarithm.

2 The construction related to the Legendre symbol

Let $p > 2$ be a prime and $f \in \mathbb{F}_p[x]$ a squarefree polynomial of positive degree d . Define the binary sequence $(\ell_n^{(p)})_{n \geq 0}$ by

$$\ell_n^{(p)} = \begin{cases} 1, & \text{if } \left(\frac{f(n)}{p}\right) = 1, \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

where $\left(\frac{\cdot}{p}\right)$ denotes the Legendre symbol modulo p . When $f(n) = n$, this reduces to the classical Legendre sequence.

Lemma 1 Suppose f has no multiple zeros in the algebraic closure of $\mathbb{F}_p$, and

- (i) $k < p$ and 2 is a primitive root modulo p , or
- (ii) $(4k)^d < p$.

Then for $V \subseteq \{0, 1, \dots, k-1\}$, we have

$$\left| \sum_{i=0}^{p-1} \prod_{j \in V} \left(\frac{f(i-j)}{p} \right) \right| \ll kdp^{\frac{1}{2}}.$$

Proof The proof is similar to that of [11, Theorem 1]. □

In this study, we focus on the arithmetic crosscorrelation between two sequence instances belonging to the same algebraic family (as specified in Eq. (2)), ensuring that both sequences are of the same degree. This symmetric setting enables a direct comparison between two algebraic objects of equal complexity, performed over the relatively prime periods p and r .

Lemma 2 Let p and r be distinct odd primes with $2 < p < r$, and let $f \in \mathbb{F}_p[x]$ and $g \in \mathbb{F}_r[x]$ be squarefree polynomials of degree d with no multiple zeros in the algebraic closures of $\mathbb{F}_p$ and $\mathbb{F}_r$, respectively. Define the binary sequences

$$s_i = \begin{cases} 1, & \text{if } \left(\frac{f(i)}{p}\right) = 1, \\ 0, & \text{otherwise,} \end{cases}$$

$$t_i = \begin{cases} 1, & \text{if } \left(\frac{g(i)}{r}\right) = 1, \\ 0, & \text{otherwise,} \end{cases}$$

of periods p and r in (2), and satisfies the condition in Lemma 1. Then for any integer $k \geq 0$ and any pattern $\underline{e} \in \{0, 1\}^{2k+2}$, the number

$$\Sigma_k(\underline{e}) = \# \{i = 0, 1, \dots, pr-1 : (s_{i-k}, \dots, s_i, t_{i-k}, \dots, t_i) = \underline{e}\}$$

satisfies

$$\Sigma_k(e) = \frac{pr}{2^{2k+2}} + O\left(2^{-k} kdp^{1/2} r\right).$$

for $k \leq \min \left\{ \frac{0.5 \log p}{\log 2}, \frac{0.5 \log r - \log \log r - \log \log p}{\log 2} \right\}$.

In particular, if $d = 1$ for classical Legendre sequences, then

$$\Sigma_k(e) = \frac{pr}{2^{2k+2}} + O\left(2^{-k} k p^{1/2} r\right).$$

Proof Write $\underline{e} = (e_0, \dots, e_{2k+1})$, for i and $j \leq k$ with $p \nmid f(i-j)$ and $r \nmid g(i-j)$, we have

$$\mathbf{1}_{\{(s_{i-j}, t_{i-j}) = (e_{k-j}, e_{2k+1-j})\}} = \frac{1 - (-1)^{e_{k-j}} \left(\frac{f(i-j)}{p} \right)}{2} \cdot \frac{1 - (-1)^{e_{2k+1-j}} \left(\frac{g(i-j)}{r} \right)}{2}$$

, where $\mathbf{1}$ is the indicator function. And thus

$$\Sigma_k = \sum_{i=0}^{pr-1} \prod_{j=0}^k \frac{1 - (-1)^{e_{k-j}} \left(\frac{f(i-j)}{p} \right)}{2} \cdot \frac{1 - (-1)^{e_{2k+1-j}} \left(\frac{g(i-j)}{r} \right)}{2} + O(kd(r+p)),$$

where the O -term arises from those indices i for which there exists some $j \in \{0, 1, \dots, k\}$ such that $\gcd(f(i-j), p) > 1$ or $\gcd(g(i-j), r) > 1$. Carrying out the expansion, we derive

$$\Sigma_k = \frac{1}{2^{2k+2}} \sum_{U, V \subseteq \{0, 1, \dots, k\}} \left(\prod_{j \in U} (-1)^{e_{k-j}} \right) \left(\prod_{j \in V} (-1)^{e_{2k+1-j}} \right) S(U, V) + O(kd(r+p)),$$

where $S(U, V) = \sum_{i=0}^{pr-1} \prod_{j \in U} \left(\frac{f(i-j)}{p} \right) \prod_{j \in V} \left(\frac{g(i-j)}{r} \right)$.

case(i) $U = V = \emptyset$, $S(\emptyset, \emptyset) = pr$, the contribution to Σ_k is $\frac{pr}{2^{2k+2}}$.

case(ii) $U = \emptyset$ and $V \neq \emptyset$, $S(\emptyset, V) = p \cdot \sum_{b=0}^{r-1} \prod_{j \in V} \left(\frac{g(b-j)}{r} \right) = O(kdpr^{\frac{1}{2}})$ by Lemma 1. Since there are $2^{k+1} - 1$ instances where $V \neq \emptyset$, the total contribution of this part to Σ_k is bounded by $O(2^{-k} k d p r^{\frac{1}{2}})$.

case(iii) $V = \emptyset$ and $U \neq \emptyset$, likewise, the cases where $V = \emptyset$ (which encompasses $2^{k+1} - 1$ choices for $U \neq \emptyset$) contribute $O(2^{-k} k d p^{\frac{1}{2}} r)$ to Σ_k .

case(iv) Meanwhile, the remaining $2^{2k+2} - 2^{k+2} + 1$ pairs (U, V) with $U \neq \emptyset$ and $V \neq \emptyset$ have a contribution bounded by

$$\frac{1}{2^{2k+2}} (2^{2k+2} - 2^{k+2} + 1) \times O(k^2 d^2 (pr)^{\frac{1}{2}}) = O(k^2 d^2 (pr)^{\frac{1}{2}}).$$

by the Chinese Remainder Theorem and Lemma 1.

Combining these results and after verifying that

$$\max\{kd(r+p), 2^{-k} k d p r^{\frac{1}{2}}, 2^{-k} k d p^{\frac{1}{2}} r, k^2 d^2 (pr)^{\frac{1}{2}}\} = 2^{-k} k d p^{\frac{1}{2}} r$$

holds for $k \leq \min \left\{ \frac{0.5 \log p}{\log 2}, \frac{0.5 \log r - \log \log r - \log \log p}{\log 2} \right\}$, we arrive at the desired conclusion. $\square$

Lemma 3 Let p and r be distinct odd primes with $2 < p < r$, $K = \min \left\{ \frac{0.5 \log p}{\log 2}, \frac{0.5 \log r - \log \log r - \log \log p}{\log 2} \right\}$, and let $M = \lfloor K \rfloor$, then $2^{-M} = O(p^{-\frac{1}{2}} (\log p)^2)$.

Proof Since $M = \lfloor K \rfloor$, we have $K - 1 < M \leq K$, then $2^{-K} \leq 2^{-M} < 2 \times 2^{-K}$. Let $A = \frac{0.5 \log p}{\log 2}$ and $B = \frac{0.5 \log r - \log \log r - \log \log p}{\log 2}$.

case(i) If $A \leq B$, then $2^{-K} = 2^{-A} = p^{-\frac{1}{2}}$.

case(ii) If $B < A$, let $x = \log r$, $y = \log p$, we have $x - y \leq 2(\log x + \log y)$. Since $\log x < \frac{2x}{5}$, we have $y < x \leq 5(y + 2 \log y)$, which implies $\log r = O(\log p)$, that is, there exists a constant $c > 0$ such that $\log r < c \log p$. Then $2^{-K} = 2^{-B} = r^{-\frac{1}{2}} \log r \log p = p^{-\frac{1}{2}} \left(\frac{p}{r}\right)^{\frac{1}{2}} \log r \log p \leq c \cdot p^{-\frac{1}{2}} (\log p)^2$.

The proof is complete. $\square$

Theorem 1 Let $\mathcal{S} = (s_i)_{i \geq 0}$ and $\mathcal{T} = (t_i)_{i \geq 0}$ be two binary sequences of prime periods p and r defined by (2). For $2 < p < r$, if $d < \frac{\log p}{\log \log p + \log(2/\log 2)}$ or 2 is a primitive root modulo p and q , the arithmetic crosscorrelation $C_{\mathcal{S}, \mathcal{T}}^A(\tau)$ satisfies

$$C_{\mathcal{S}, \mathcal{T}}^A(\tau) \ll dp^{\frac{1}{2}} r (\log p)^2, \quad 0 \leq \tau < pr.$$

Proof By [8, Theorem 1], the arithmetic crosscorrelation of two binary sequences with coprime periods is constant. Therefore, we may set $\tau = 0$. Furthermore, without loss of generality, we can assume $S(2) \geq T^{(0)}(2)$, where $S(2) = \sum_{i=0}^{pr-1} s_i 2^i$, $T^{(0)}(2) = \sum_{i=0}^{pr-1} t_i 2^i$. We write

$$S(2) - T^{(0)}(2) = \sum_{i=0}^{pr-1} (s_i - t_i) 2^i = \sum_{i=0}^{pr-1} w_i 2^i$$

with $w_i \in \{0, 1\}$. Since the values w_i are unique, it is necessary to estimate the number of indices i for which $w_i = 1$.

Assume there exist integers $n \geq k \geq 1$ and $a \in \{0, 1\}$ such that

$$\begin{pmatrix} s_{n-k} & s_{n-k+1} & s_{n-k+2} & \cdots & s_{n-1} & s_n \\ t_{n-k} & t_{n-k+1} & t_{n-k+2} & \cdots & t_{n-1} & t_n \end{pmatrix} = \begin{pmatrix} a & x_1 & x_2 & \cdots & x_{k-1} & y \\ 1-a & x_1 & x_2 & \cdots & x_{k-1} & z \end{pmatrix}, \quad y, z \in \{0, 1\},$$

For $a = 1$, we have

$$w_n = \begin{cases} 1, & \text{if } y \neq z, \\ 0, & \text{otherwise.} \end{cases}$$

For $a = 0$, we have

$$w_n = \begin{cases} 1, & \text{if } y = z, \\ 0, & \text{otherwise.} \end{cases}$$

In total, there exist 2^{k+1} distinct patterns $\underline{e} \in \{0, 1\}^{2k+2}$ such that

$$(s_{n-k}, s_{n-k+1}, \dots, s_n, t_{n-k}, t_{n-k+1}, \dots, t_n) = \underline{e}$$

implies $w_n = 1$. According to Lemma 2, for every $k = 1, 2, \dots$, there exist at least

$$2^{k+1} \frac{pr}{2^{2k+2}} + O\left(\frac{2^{k+1}}{2^k} kdp^{\frac{1}{2}} r\right) = \frac{pr}{2^{k+1}} + O(kdp^{\frac{1}{2}} r)$$

distinct integers n in every fixed interval of length pr such that $w_n = 1$.

Choose $M = \left\lfloor \min \left\{ \frac{0.5 \log p}{\log 2}, \frac{0.5 \log r - \log \log r - \log \log p}{\log 2} \right\} \right\rfloor \leq \log p$. By Lemma 3, for $n \in [0, pr - 1]$ with $w_n = 1$, we have the number N_1

$$\begin{aligned} N_1 &\geq \sum_{k=1}^M \frac{pr}{2^{k+1}} + O\left(\sum_{k=1}^M kdp^{\frac{1}{2}}r\right) \\ &= \frac{pr}{2} \left(1 - \left(\frac{1}{2}\right)^M\right) + O(M^2 dp^{\frac{1}{2}}r) \\ &= \frac{pr}{2} + O(dp^{\frac{1}{2}}r(\log p)^2). \end{aligned}$$

Analogously, the number N_0 for which $w_n = 0$ satisfies $N_0 \geq \frac{pr}{2} + O(dp^{\frac{1}{2}}r(\log p)^2)$. Then the result yields by (1). $\square$

Remark 1 Let $f(x) = g(x) = x$, this is then the Legendre sequence, which is consistent with the result of [8, Theorem 2].

3 The construction related to the trace function

Let $q = 2^k$ with $q - 1$ is prime. Let $f \in \mathbb{F}_q[x]$ be a polynomial of odd degree $d \geq \log q$ such that for any term ax^e of f , the coefficient a vanishes if and only if e is even. Denote by Tr the absolute trace function from $\mathbb{F}_{2^k}$ to $\mathbb{F}_2$, and let α be a primitive element of $\mathbb{F}_{2^k}$. Define a sequence $(d_n)_{n \geq 0}$ by

$$d_n = \text{Tr}_k(f(\alpha^n)), \quad n = 0, 1, 2, \dots \quad (3)$$

This sequence is periodic with period $q - 1$.

Similarly, we only consider the arithmetic crosscorrelation when the polynomial degrees are the same. And note $\gcd(n_1, n_2) = 1$ if and only if $\gcd(2^{n_1} - 1, 2^{n_2} - 1) = 1$.

Lemma 4 Let $p = 2^{n_1} - 1$ and $r = 2^{n_2} - 1$ be two distinct primes, where $n_1 < n_2$ are coprime. Let $\mathcal{S} = (s_n)_{n \geq 0}$ and $\mathcal{T} = (t_n)_{n \geq 0}$ be the p -periodic and r -periodic sequences respectively, defined by

$$s_n = \text{Tr}_{n_1}(f(\alpha_1^n)), \quad t_n = \text{Tr}_{n_2}(g(\alpha_2^n)), \quad n \geq 0,$$

where the polynomial f and g have the same positive odd degree $d \geq \log 2^{n_2}$ defined in (3).

For an integer $k \geq 1$ and any pattern $\underline{e} \in \{0, 1\}^{2k+2}$, the number Σ_k of indices $i = 0, 1, \dots, pr - 1$ such that

$$(s_{i-k}, s_{i-k+1}, \dots, s_i, t_{i-k}, t_{i-k+1}, \dots, t_i) = \underline{e}$$

satisfies

$$\Sigma_k(\underline{e}) = \frac{pr}{2^{2k+2}} + O\left(2^{-k} dpr^{\frac{1}{2}} + 2^{-k} dp^{\frac{1}{2}}r + d^2(pr)^{\frac{1}{2}}\right).$$

Proof Define the additive character $\psi_{n_i}(x) = (-1)^{\text{Tr}_{n_i}(x)}$ for $i = 1, 2$. Write $\underline{e} = (e_0, \dots, e_{2k+1})$, for i and $j \leq k$, we have

$$\mathbf{1}_{\{(s_{i-j}, t_{i-j}) = (e_{k-j}, e_{2k+1-j})\}} = \frac{1 + (-1)^{e_{k-j}} (\psi_{n_1}(f(\alpha_1^{i-j})))}{2} \cdot \frac{1 + (-1)^{e_{2k+1-j}} (\psi_{n_2}(g(\alpha_2^{i-j})))}{2}$$

, where $\mathbf{1}$ is the indicator function. And thus

$$\Sigma_k = \sum_{i=0}^{pr-1} \prod_{j=0}^k \frac{1 + (-1)^{e_{k-j}}(\psi_{n_1}(f(\alpha_1^{i-j})))}{2} \cdot \frac{1 + (-1)^{e_{2k+1-j}}(\psi_{n_2}(g(\alpha_2^{i-j})))}{2},$$

Carrying out the expansion, we derive

$$\Sigma_k = \frac{1}{2^{2k+2}} \sum_{U, V \subseteq \{0, 1, \dots, k\}} \left(\prod_{j \in U} (-1)^{e_{k-j}} \right) \left(\prod_{j \in V} (-1)^{e_{2k+1-j}} \right) S(U, V),$$

where $S(U, V) = \sum_{i=0}^{pr-1} \prod_{j \in U} \psi_{n_1}(f(\alpha_1^{i-j})) \prod_{j \in V} \psi_{n_2}(g(\alpha_2^{i-j}))$.

case(i) $U = V = \emptyset$, $S(\emptyset, \emptyset) = pr$, the contribution to Σ_k is $\frac{pr}{2^{2k+2}}$.

case(ii) $U = \emptyset$ and $V \neq \emptyset$, $S(\emptyset, V) = p \cdot \sum_{b=0}^{r-1} \prod_{j \in V} \psi_{n_2}(g(\alpha_2^{i-j})) = O(dpr^{\frac{1}{2}})$ by [12, Theorem 5.38]. Since there are $2^{k+1} - 1$ instances where $V \neq \emptyset$, the total contribution of this part to Σ_k is bounded by $O(2^{-k} dpr^{\frac{1}{2}})$.

case(iii) $V = \emptyset$ and $U \neq \emptyset$, likewise, the cases where $V = \emptyset$ (which encompasses $2^{k+1} - 1$ choices for $U \neq \emptyset$) contribute $O(2^{-k} dp^{\frac{1}{2}} r)$ to Σ_k .

case(iv) Meanwhile, the remaining $2^{2k+2} - 2^{k+2} + 1$ pairs (U, V) with $U \neq \emptyset$ and $V \neq \emptyset$ have a contribution bounded by

$$\frac{1}{2^{2k+2}} (2^{2k+2} - 2^{k+2} + 1) \times O(d^2(pr)^{\frac{1}{2}}) = O(d^2(pr)^{\frac{1}{2}}).$$

by the Chinese Remainder Theorem.

Combining these results, we arrive at the desired conclusion. $\square$

Theorem 2 Let $\mathcal{S} = (s_i)_{i \geq 0}$ and $\mathcal{T} = (t_i)_{i \geq 0}$ be two binary sequences of prime periods p and r defined by (3). For $n_1 < n_2$ and $\gcd(n_1, n_2) = 1$, if $d > \log 2^{n_2}$, the arithmetic crosscorrelation $C_{\mathcal{S}, \mathcal{T}}^A(\tau)$ satisfies

$$C_{\mathcal{S}, \mathcal{T}}^A(\tau) \ll \max \left\{ dp^{\frac{1}{2}} r \log p, d^2 pr^{\frac{1}{2}} \right\}, \quad 0 \leq \tau < pr.$$

Proof As in the proof of Theorem 1 and according to Lemma 4, for every $k = 1, 2, \dots$, there exist at least

$$\frac{pr}{2^{k+1}} + O\left(dpr^{\frac{1}{2}} + dp^{\frac{1}{2}} r + 2^k d^2(pr)^{\frac{1}{2}}\right)$$

distinct integers n in every fixed interval of length pr such that $w_n = 1$.

Choose $M = \left\lfloor \frac{0.5 \log p}{\log 2} \right\rfloor$. For $n \in [0, pr - 1]$ with $w_n = 1$, we have the number N_1

$$\begin{aligned} N_1 &\geq \sum_{k=1}^M \frac{pr}{2^{k+1}} + \sum_{k=1}^M O\left(dpr^{\frac{1}{2}} + dp^{\frac{1}{2}} r + 2^k d^2(pr)^{\frac{1}{2}}\right) \\ &= \frac{pr}{2} \left(1 - \left(\frac{1}{2}\right)^M\right) + O(Mdpr^{\frac{1}{2}} + Mdp^{\frac{1}{2}} r + 2^{M+1} d^2(pr)^{\frac{1}{2}}) \\ &= \frac{pr}{2} + O(dp^{\frac{1}{2}} r \log p + d^2 pr^{\frac{1}{2}}), \end{aligned}$$

since $p^{-\frac{1}{2}} \leq 2^{-M} < 2p^{-\frac{1}{2}}$, $(\frac{1}{2} \log_2 p - 1) dp^{\frac{1}{2}} r < Mdp^{\frac{1}{2}} r \leq \frac{1}{2} (\log_2 p) dp^{\frac{1}{2}} r$ and $\frac{1}{2} d^2 pr^{\frac{1}{2}} < 2^M d^2(pr)^{\frac{1}{2}} \leq d^2 pr^{\frac{1}{2}}$.

Analogously, the number N_0 for which $w_n = 0$ satisfies $N_0 \geq \frac{pr}{2} + O(dp^{\frac{1}{2}}r \log p + d^2 pr^{\frac{1}{2}})$.

Then the result yields by (1). $\square$

4 Arithmetic crosscorrelation of non-binary sequences of coprime period

While Theorem 1 of [8] establishes the constancy of the arithmetic crosscorrelation for the binary coprime period case, the result in fact extends to non-binary sequences as well.

Let $\mathcal{S} = (s_i)_{i \geq 0}$ and $\mathcal{T} = (t_i)_{i \geq 0}$ be two periodic sequences over $\mathbb{Z}_m = \{0, 1, \dots, m-1\}$ with periods p and q , respectively, and $\gcd(p, q) = 1$. Denote $N = pq$, the common period of $\mathcal{S}$ and $\mathcal{T}$. Let $\zeta_m = e^{2\pi\sqrt{-1}/m}$. For a fixed integer τ , define the arithmetic crosscorrelation $C_{\mathcal{S}, \mathcal{T}}^A(\tau)$ of $\mathcal{S}$ and $\mathcal{T}$ as follows. Represent

$$\alpha = \sum_{i=0}^{\infty} s_i m^i, \quad \beta_{\tau} = \sum_{i=0}^{\infty} t_{i+\tau} m^i,$$

and let $\alpha - \beta_{\tau} = \sum_{i=0}^{\infty} u_i m^i$ be the (eventually periodic) m -adic expansion. Let j be the length of the preperiod. Then

$$C_{\mathcal{S}, \mathcal{T}}^A(\tau) = \sum_{i=j}^{j+N-1} \zeta_m^{u_i} = \sum_{j=0}^{m-1} M_j \zeta_m^j, \quad (4)$$

where M_j counts the times j appears in one full period of the m -adic expansion of $\alpha - \beta_{\tau}$.

Theorem 3 *Let $m \geq 2$ be an integer. Let $\mathcal{S}$ and $\mathcal{T}$ be two periodic sequences over $\mathbb{Z}_m$ with coprime periods p and q , respectively. Then the arithmetic crosscorrelation $C_{\mathcal{S}, \mathcal{T}}^A(\tau)$ is independent of τ .*

Proof Write $N = pq$. We define the first N coefficients of the periodic part of the difference sequence between sequence $\mathcal{S}$ and sequence $\mathcal{T}$ as

$$\Delta_{\tau} = \sum_{i=0}^{N-1} \omega_i m^i.$$

for $\omega_i \in \{0, 1, \dots, m-1\}$.

Let $S(m) = \sum_{i=0}^{N-1} s_i m^i$ and $T^{(\tau)}(m) = \sum_{i=0}^{N-1} t_{i+\tau} m^i$, according the Proposition 13.3.8 in [2] and an analysis analogous to that in [6], one can check that either

$$S(m) - T^{(\tau)}(m) = \Delta_{\tau}$$

with $S(m) \geq T^{(\tau)}(m)$, otherwise

$$m^N - 1 + (S(m) - T^{(\tau)}(m)) = \Delta_{\tau}.$$

Hence the N -tuple $(u_j, \dots, u_{j+N-1})$ is uniquely determined by Δ_τ itself (as an integer between 0 and $m^N - 2$).

Now fix τ and let x be the unique integer $0 \leq x < q$ such that

$$x \equiv \tau p^{-1} \pmod{q},$$

which exists because $\gcd(p, q) = 1$. Set

$$\lambda_k = \sum_{i=0}^{p-1} (s_i - t_{i+kp}) m^i, \quad k = 0, 1, \dots, q-1.$$

Because $\mathcal{T}$ has period q , we have $\lambda_{k+q} = \lambda_k$. Then

$$m^{xp} \Delta_\tau = \sum_{n=0}^{N-1} (s_n - t_{n+\tau}) m^{n+xp}.$$

Write $n = n_1 + n_2 p$ with $0 \leq n_1 < p$, $0 \leq n_2 < q$. Let $n_3 \equiv x + n_2 \pmod{q}$ with $0 \leq n_3 < q$. Using the periodicity of $\mathcal{S}$ and $\mathcal{T}$, we obtain

$$m^{xp} \Delta_\tau \equiv \sum_{n_3=0}^{q-1} \lambda_{n_3} m^{n_3 p} \pmod{m^N - 1}. \quad (5)$$

Denote

$$\Omega = \sum_{n_3=0}^{q-1} \lambda_{n_3} m^{n_3 p} \pmod{m^N - 1},$$

notice that Ω does not depend on τ .

Now consider the m -adic expansion sequences. Multiplication by m^{xp} modulo $m^N - 1$ corresponds to a cyclic right shift of the N coefficients m -adic representation by xp positions. This shift does not alter the multiset of digits in one full period. Eq. (5) therefore implies that the sequence corresponding to Δ_τ is a cyclic shift of the sequence corresponding to Ω . Consequently, for every τ , the number of occurrences of each $j \in \mathbb{Z}_m$ in the N coefficients period of Δ_τ is exactly the same as in Ω .

Then by (4), the proof is completed. $\square$

5 Conclusions

In this paper, we studied the arithmetic crosscorrelation of sequences with coprime periods. For sequences constructed from the Legendre symbol, which generalize the classical Legendre sequences, we established an upper bound of $O(dp^{1/2}r(\log p)^2)$ for distinct prime periods $p < r$. For sequences generated by trace functions, we obtained a bound of $O(\max(dp^{\frac{1}{2}}r \log p, d^2 pr^{\frac{1}{2}}))$ for periods $p = 2^{n_1} - 1$ and $r = 2^{n_2} - 1$ with $\gcd(n_1, n_2) = 1$.

Furthermore, we extended the constant property of arithmetic crosscorrelation to the non-binary case, proving that for any two sequences with coprime periods, the arithmetic crosscorrelation is independent of the shift.

Future work may explore the arithmetic crosscorrelation for other families of sequences with coprime periods, such as non-binary Legendre sequences and non-binary m -sequences.

Declarations

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- Conflict of interest
The authors declare no competing interests.

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